

NirogScan Comprehensive Health Report

Patient Information

Name: ak2
Age: 24
Gender: Male
Weight: N/A
Height: N/A
Notes: None

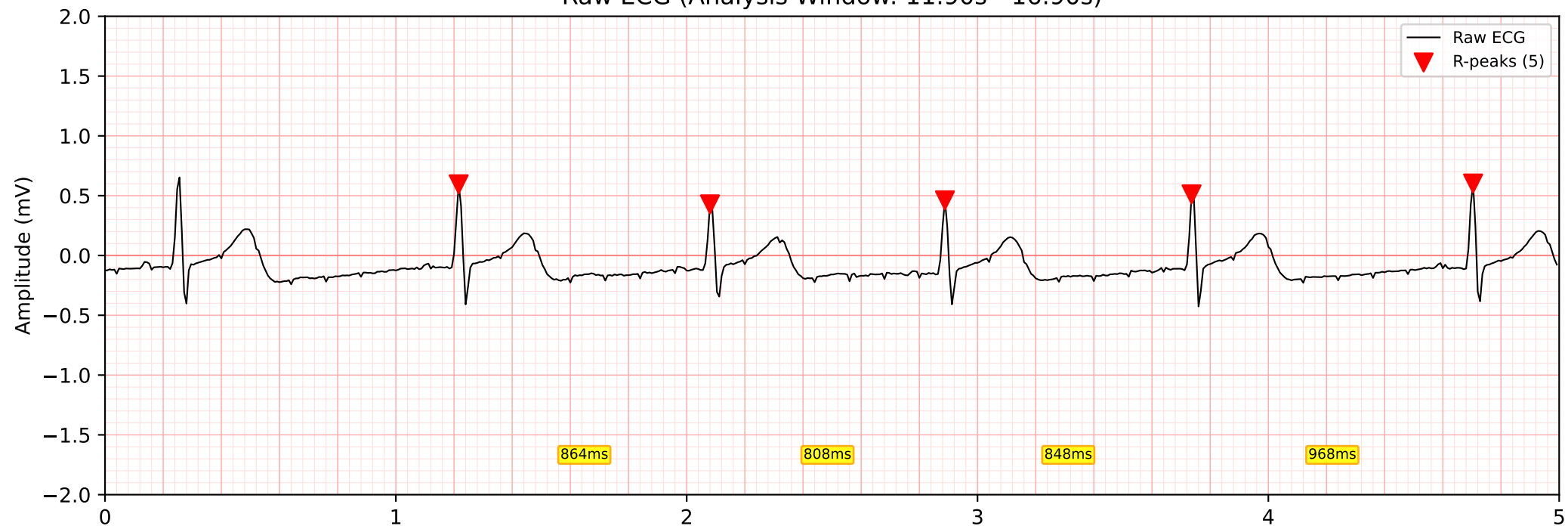
Recording & Analysis Summary

ECG Duration: 21.6s (2705 samples)
ECG Sampling Rate: 125 Hz
Best Analysis Window: 11.90s - 16.90s
Window SNR: 21.8 dB
R-peaks Detected: 5
Analysis Engine: NeuroKit2 v0.2.12
ECG Heart Rate: 69.1 ± 4.5 bpm
PPG Duration: 43.3s

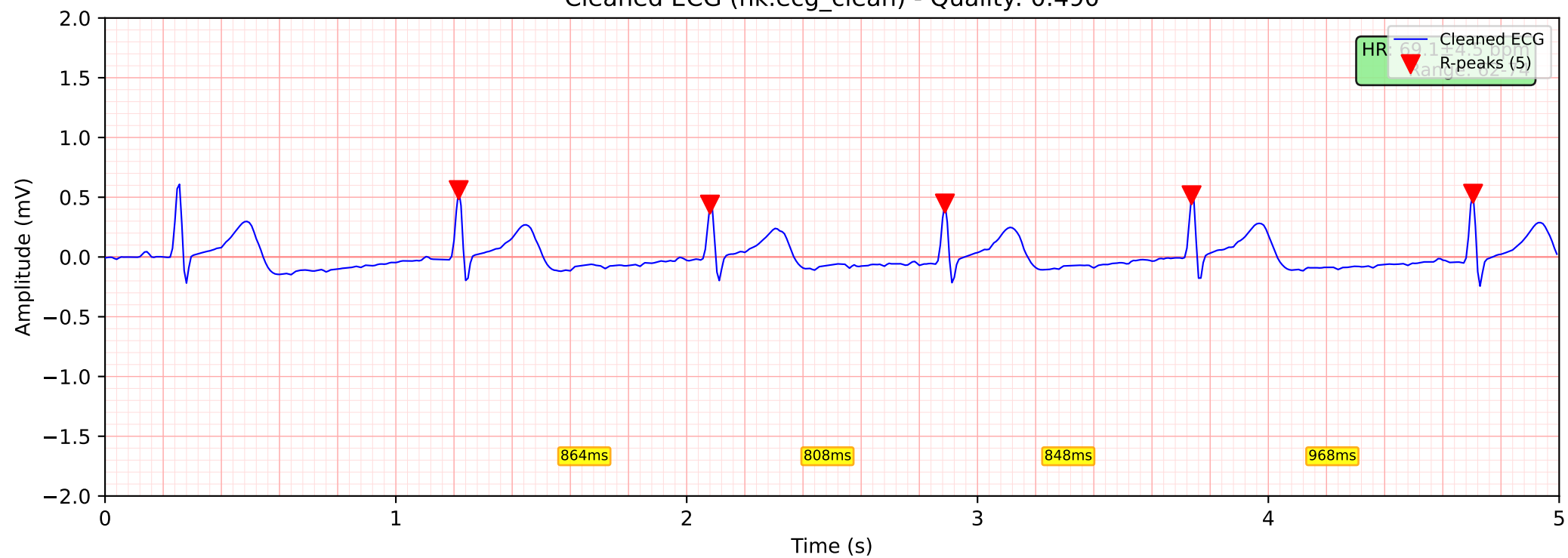
Report Generated: 2026-01-31 19:27:13

ECG Signal Analysis

Raw ECG (Analysis Window: 11.90s - 16.90s)

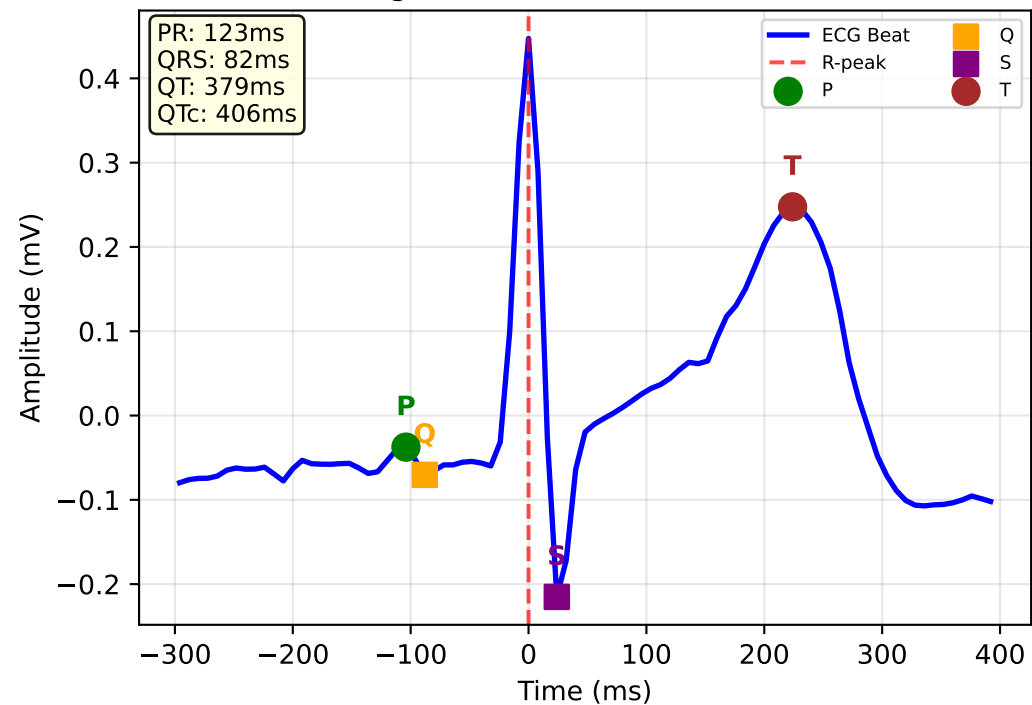


Cleaned ECG (nk.ecg_clean) - Quality: 0.490

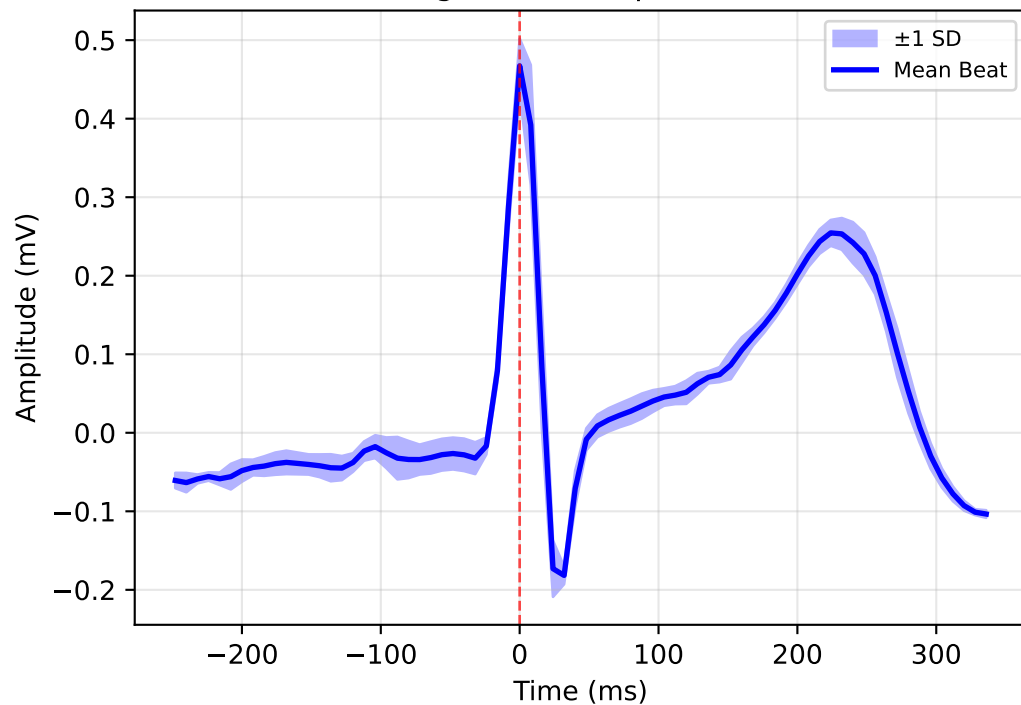


ECG Morphological Analysis (PQRST Detection)

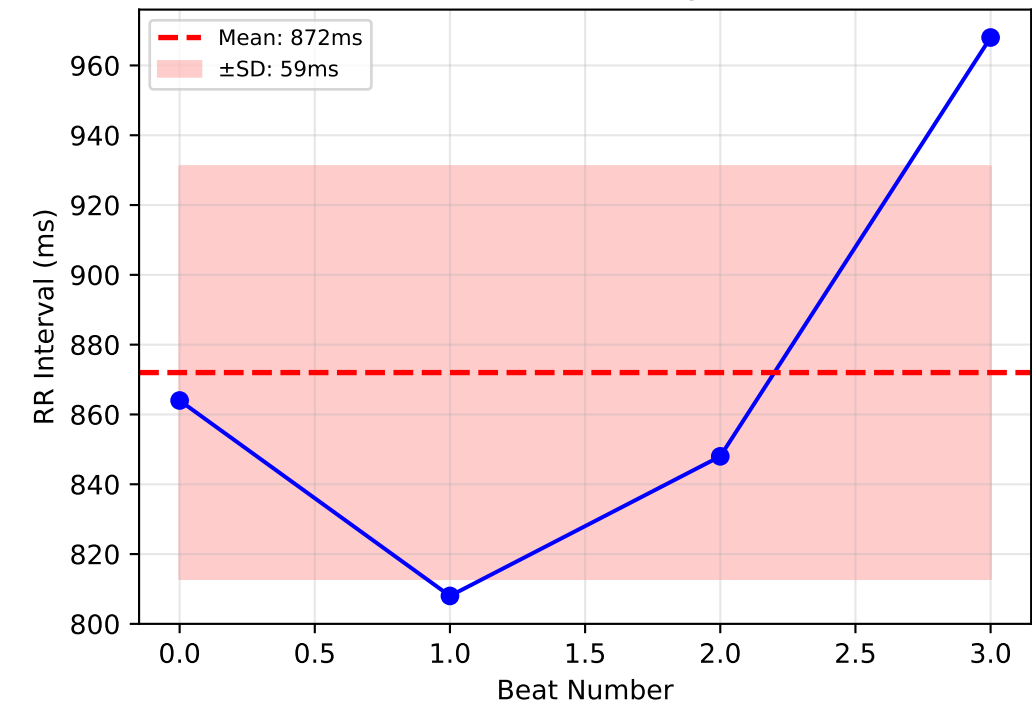
Single Beat with PQRST Waves



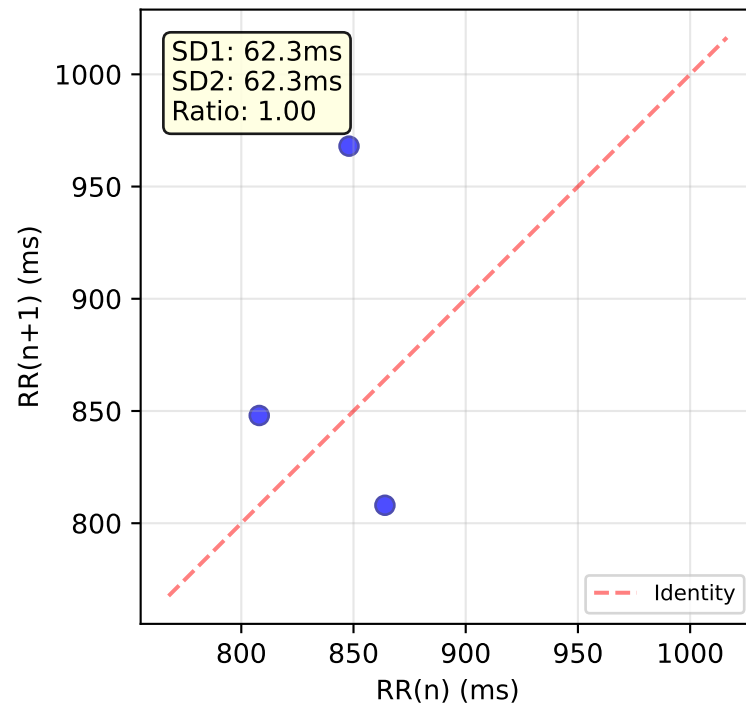
Average Beat Template (n=3)



RR Interval Tachogram



Poincaré Plot



HRV Analysis - Time Domain

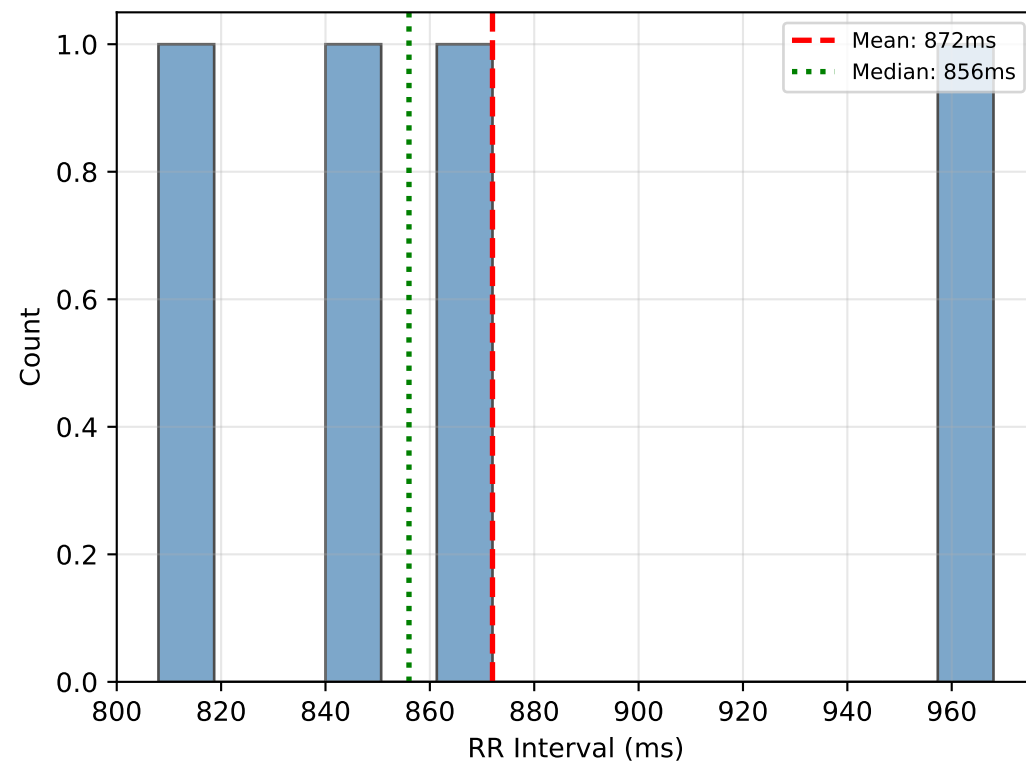
Core Time Domain Metrics

Average RR Interval:	872.00
Heart Rate Variability:	68.20
Short-term HRV:	79.87
Percentage of Large Changes:	50.00
Percentage of Small Changes:	75.00
Successive Differences SD:	88.12

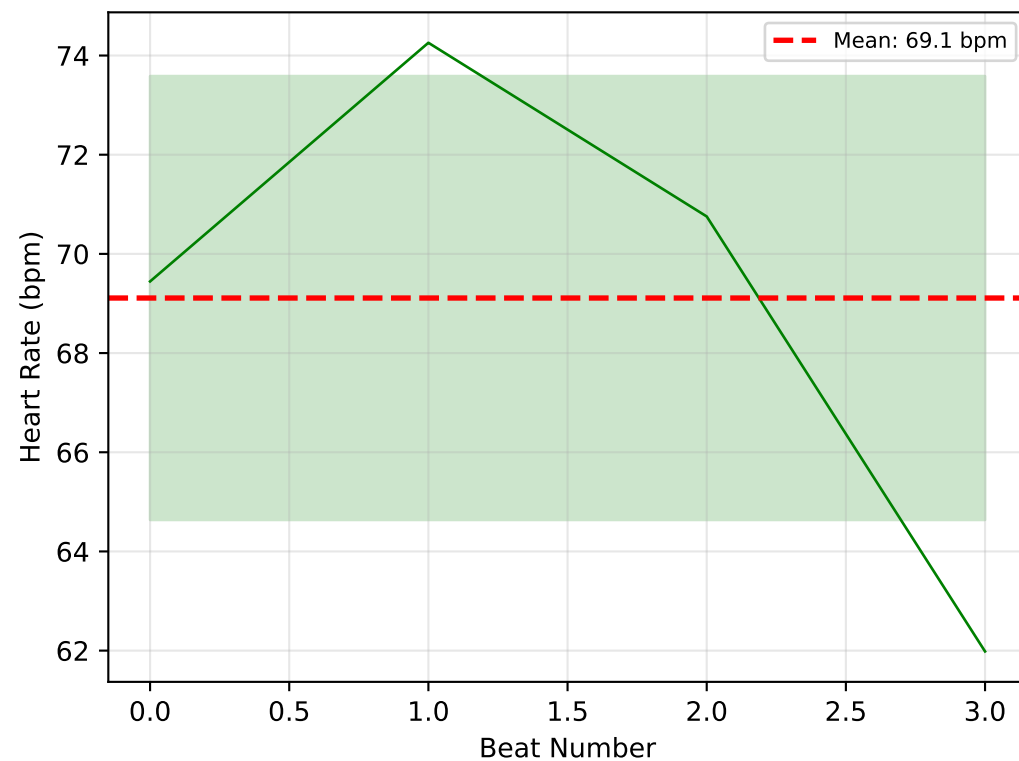
Additional Time Metrics

Median RR Interval:	856.00
Median Absolute Deviation:	41.51
Coefficient of Variation:	0.08
Interquartile Range:	52.00
Triangular Index:	0.00
HRV Triangular Index:	4.00

RR Interval Distribution



Instantaneous Heart Rate



HRV Analysis - Frequency & Nonlinear Domains

Frequency Domain Metrics

HRV_TP: 0.00

Insufficient frequency data

Nonlinear Metrics (Poincaré)

Poincaré SD1: 62.3110

Poincaré SD2: 62.3110

SD1/SD2 Ratio: 1.0000

Poincaré Area: 12197.7571

Cardiac Sympathetic Index: 1.0000

Cardiac Vagal Index: 4.7933

Entropy & Fractal Metrics

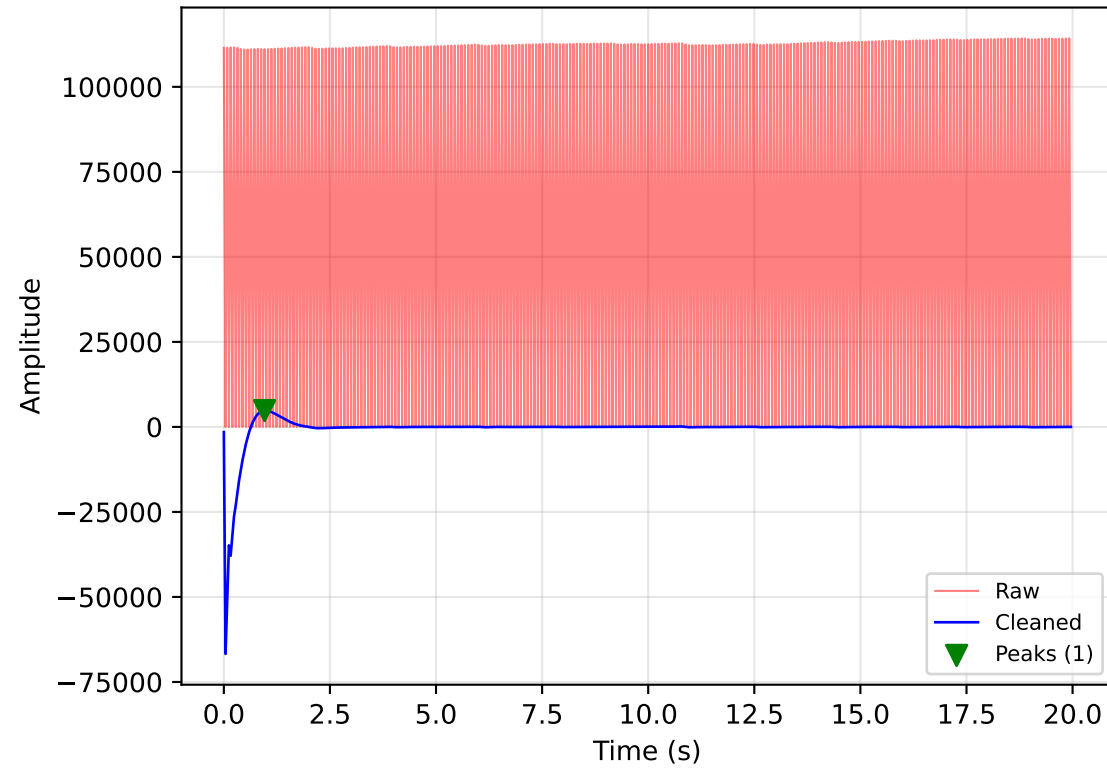
Approximate Entropy: 0.4055

Sample Entropy: -inf

Fuzzy Entropy: 3.1282

PPG Signal Analysis (MAX30102 IR Channel)

PPG Signal with Peak Detection

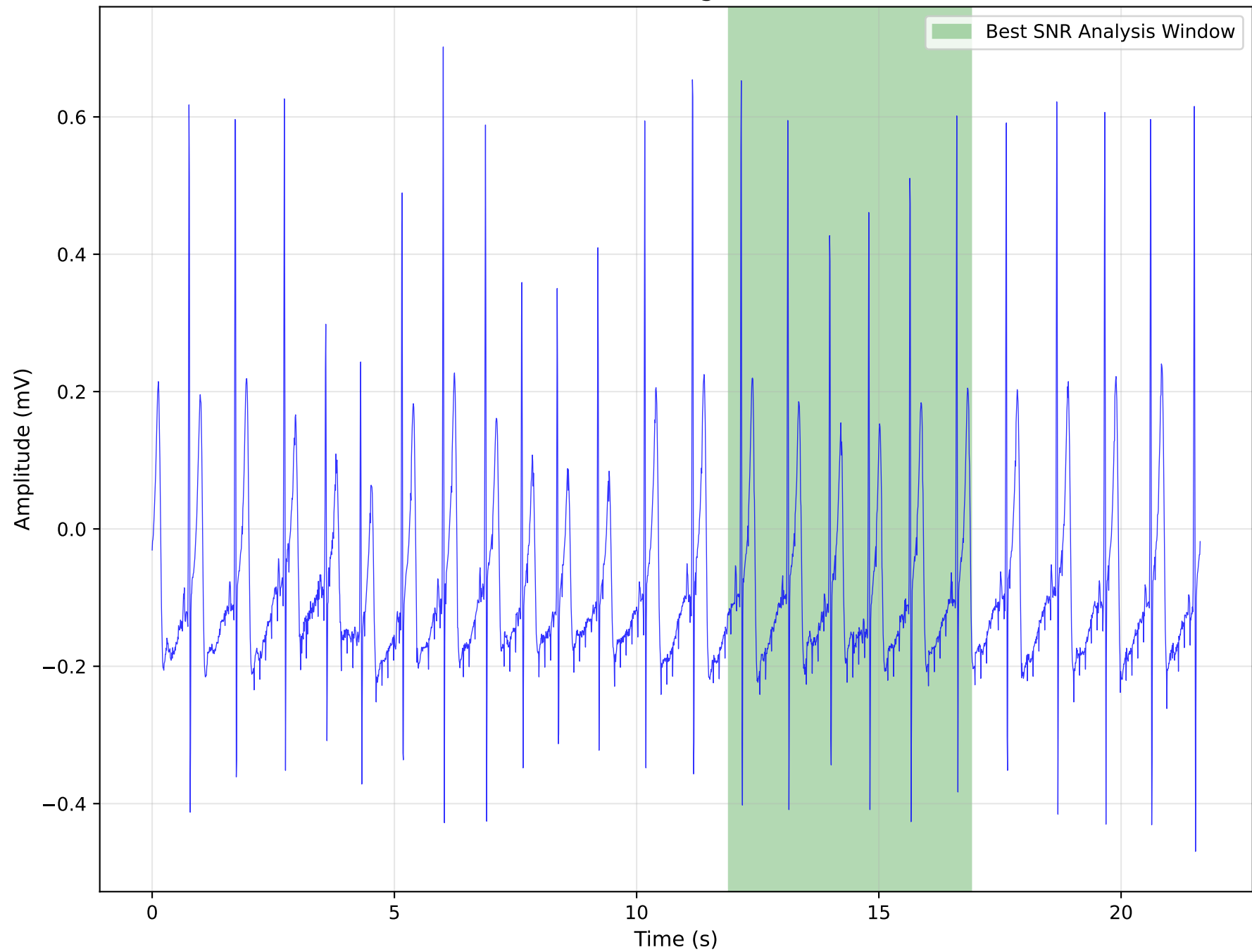


Insufficient HR data

Insufficient data

Insufficient data

Full ECG Recording (21.6s total)



Analysis Summary

ECG Analysis:

Signal Quality (SNR):	21.8 dB
R-peaks Detected:	5
Heart Rate:	69.1 ± 4.5 bpm
SDNN:	68.2 ms
RMSSD:	79.9 ms

PPG Analysis:

Peaks Detected:	1
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Device Information:

ECG Sensor:	AD8232 (Gain: 1100)
ECG Sampling:	125 Hz
PPG Sensor:	MAX30102 (Red 660nm, IR 880nm)
PPG Sampling:	25 Hz
ADC:	12-bit, 3300mV ref

Understanding Your Results

(What each measurement means in simple terms)

- **Average RR Interval:**

The average time between heartbeats in milliseconds. Normal range: 600-1000ms.

- **Heart Rate Variability:**

How much your heartbeat timing varies. Higher is generally better, indicating a healthy, adaptable heart. Normal: 50-100ms.

- **Short-term HRV:**

Measures beat-to-beat changes. Reflects your parasympathetic (rest & digest) nervous system activity. Normal: 20-50ms.

- **Percentage of Large Changes:**

Percentage of heartbeats that differ by more than 50ms from the previous one. Higher values indicate good heart health. Normal: 3-25%.

- **Poincaré SD1:**

Short-term variability from Poincaré plot. Related to RMSSD.

- **Poincaré SD2:**

Long-term variability from Poincaré plot. Related to SDNN.

- **Approximate Entropy:**

Signal complexity/predictability. Lower = more regular heartbeat.

- **Sample Entropy:**

Similar to ApEn but more consistent. Measures heart rhythm complexity.

- **PR Interval:**

Time from atrial to ventricular activation. Normal: 120-200ms.

- **QRS Duration:**

Time for ventricles to contract. Normal: 80-120ms.

- **QT Interval:**

Total ventricular activity time. Varies with heart rate.

- **Corrected QT:**

QT interval adjusted for heart rate. Normal: <440ms.

*Note: These measurements provide insights into heart health but are not medical diagnoses.
Please consult a healthcare professional for medical advice.*